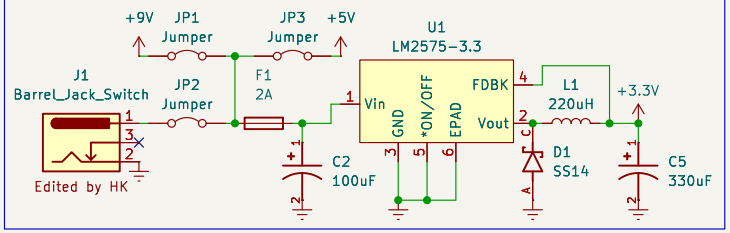
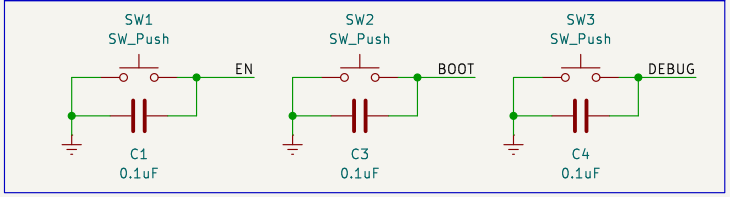
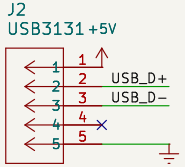
9V to 3.3V Switching Regulator:
Barrel jack will be used for testing and modularity with JP2 connected and JP1 disconnected. JP2 will be disconnected in full product and receive +9V from connector pin 1 through JP1.



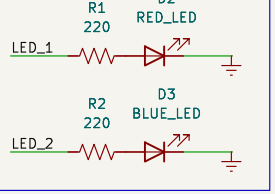
Switches:
Switch 1 turns ESP32 off, Switch 2 resets ESP32, and Switch 3 will be used to debug.

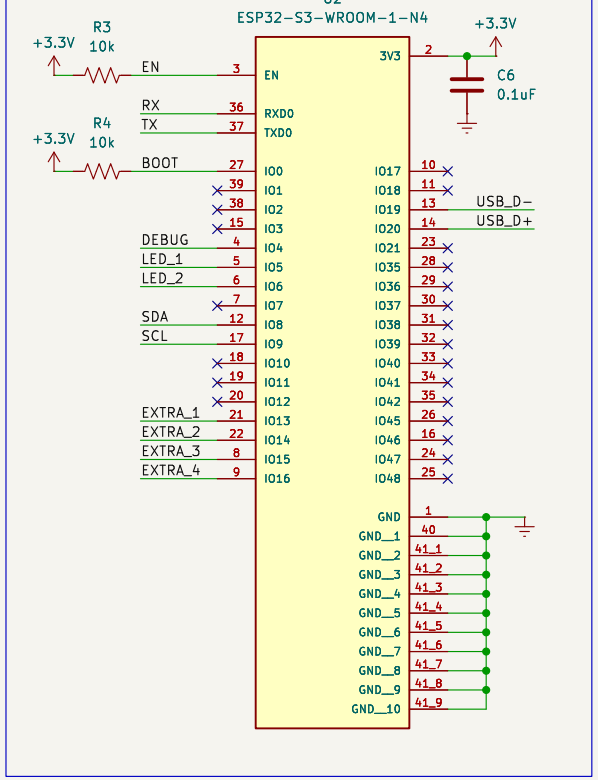


USB Port:
Used for programming SMD ESP32.

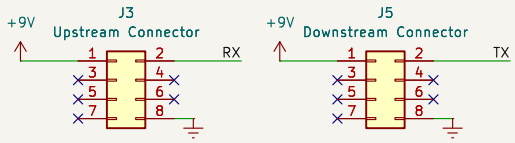


LEDs:
LED 1 will activate with debug switch.
LED 2 will activate with successful UART communication.

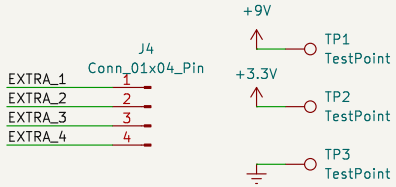




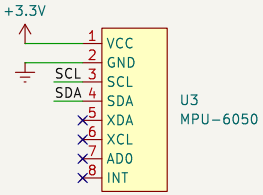
Connectors:
Power and ground is shared. UART_IN is received from subsystem B1, and UART_OUT is transmitted to subsystem C2.



Extra Pins:
Test points for all power levels are included along with extra headers in case they are needed.



Sensor:
Gyroscope and Accelerometer



All resistors are rated 1/4 W

0.1uf capacitors are rated 50V
100uf rated 25V
330uf rated 4V

Team 201

Sheet: /
File: EGR314_IndividualSchematic.kicad_sch

Title: Gyroscope & Accelerometer PCB

Size: A4	Date: 2026-02-20	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/1